

Microalgae Marvel
The Invisible Organisms Remaking India's Agricultural Future

Debabrata Sarkar
CMD, AlgaEnergy India (MASI) & President - APAC

As India Today chronicles five decades of transformation, one entrepreneur is harnessing nature's smallest powerhouses to solve agriculture's biggest challenges—turning industrial waste into farming gold.

In the sprawling fields of Punjab, where wheat waves meet industrial smokestacks, a quiet revolution is taking root—one invisible to the naked eye yet powerful enough to reshape how India feeds itself. While the past 50 years witnessed India's journey from food scarcity (PL-480 dependencies) to self-sufficiency (Green Revolution) to chemical-intensive productivity (1990s-2000s), the next chapter demands sustainable intensification. Enter Debabrata Sarkar and his audacious bet: microscopic algae as agriculture's climate solution.

At the heart of this transformation stands AlgaEnergy India , where organisms so small that five million fit in a single gram of soil are turning industrial waste into agricultural treasure while tackling climate change. It's a paradigm shift from the chemical-dependent farming that dominated India Today's coverage of agriculture across five decades—offering regeneration instead of depletion, biology instead of chemistry.

The story begins with an unconventional partnership. When global biotech giant AlgaEnergy sought an Indian ally to commercialize five decades of microalgae research, they found their match to take the Lead in Debabrata —an agriculture graduate turned IIM Kolkata alumnus who saw potential where others saw pond scum. Through a strategic joint venture with KREPL Group, AlgaEnrgy India achieved something rare in Indian agritech: profitability within three years while maintaining sustainability focus.

The science is elegantly circular. MicroAlgae captures industrial CO2 emissions—the same greenhouse gases choking our atmosphere—and feeds them to carefully cultivated microalgae. These microscopic powerhouses gorge themselves on carbon dioxide, transforming it into biomass that becomes premium biostimulants and biofertilizers. Every ton of CO2 captured becomes agricultural input, closing the loop between industry and farming.

"Traditional agriculture is like running a car on fossil fuels when we already have the technology for clean energy," explains Debabrata . "Our microalgae-based solutions don't just replace chemicals—they actively heal the soil while boosting productivity."

The results validate the promise. Field trials across agricultural universities document yield increases of 10-12% while reducing chemical fertilizer usage by up to 20%. More compelling still, soil health indicators show dramatic improvement within just one growing season—a transformation that typically requires years with conventional organic methods. Recognition has been swift: Forbes India spotlighted AlgaEnrgy India's disruptive potential, scientific journals validated environmental impact, and TEDx platforms amplified the message.

"When I see a farmer's face light up after using our products—better yields, healthier soil, reduced costs—that's when I know we're on the right path," Debabrata shares, reflecting the ground-level impact that statistics cannot capture.

As India faces the dual challenge of feeding 1.4 billion people while meeting climate commitments—pressures intensifying across the decades India Today has chronicled—AlgaEnergy India's approach offers rare convergence: industrial emissions become agricultural assets, waste transforms into wealth, and farmers emerge as climate heroes rather than environmental villains.

Debabrata's expanding footprint across Asia-Pacific signals a paradigm shift from extraction to regeneration. In an issue celebrating 50 years of documenting India's transformation, AlgaEnrgy Inida represents the next chapter: innovations so small they're invisible, yet so significant they could redefine how India—and the world—feeds itself sustainably. The invisible revolution has begun.